

```
import java.util.ArrayList;
```

```
// ===== ENCAPSULATION =====
```

```
class Book {
```

```
    private String title;
```

```
    private String author;
```

```
    private boolean available;
```

```
    public Book(String title, String author) {
```

```
        this.title = title;
```

```
        this.author = author;
```

```
        this.available = true;
```

```
    }
```

```
    public String getInfo() {
```

```
        return title + " by " + author;
```

```
    }
```

```
    public boolean isAvailable() {
```

```
        return available;
```

```
    }
```

```
    public String borrowBook() {
```

```
        if (available) {
```

```
            available = false;
```

```
            return "Book borrowed successfully.";
```

```

    }

    return "Book is not available.";
}

public String returnBook() {
    if (!available) {
        available = true;
        return "Book returned successfully.";
    }
    return "Book was not borrowed.";
}
}

// ===== ABSTRACTION =====

abstract class User {
    protected String name;

    public User(String name) {
        this.name = name;
    }

    public abstract String role();
}

// ===== INHERITANCE =====

class Student extends User {

```

```
public Student(String name) {  
    super(name);  
}
```

```
@Override  
public String role() {  
    return name + " is a Student (can borrow 2 books)";  
}  
}
```

```
class Teacher extends User {  
    public Teacher(String name) {  
        super(name);  
    }  
}
```

```
@Override  
public String role() {  
    return name + " is a Teacher (can borrow 5 books)";  
}  
}
```

```
// ===== LIBRARY CLASS =====
```

```
class Library {  
    private ArrayList<Book> books;  
  
    public Library() {
```

```

        books = new ArrayList<>();
    }

    public void addBook(Book book) {
        books.add(book);
    }

    public void showBooks() {
        System.out.println("\nAvailable Books:");
        for (Book book : books) {
            System.out.println("- " + book.getInfo() +
                (book.isAvailable() ? " (Available)" : " (Borrowed)"));
        }
    }
}

// ===== POLYMORPHISM =====

class Display {
    public static void showRoles(User[] users) {
        System.out.println("\nUser Roles:");
        for (User user : users) {
            System.out.println(user.role());
        }
    }
}

```

```
// ===== MAIN CLASS =====

public class Main {

    public static void main(String[] args) {

        // Create Library

        Library lib = new Library();

        // Add Books

        Book b1 = new Book("OOP Basics", "John Doe");
        Book b2 = new Book("Java Programming", "James Gosling");
        Book b3 = new Book("Data Structures", "Mark Allen");

        lib.addBook(b1);
        lib.addBook(b2);
        lib.addBook(b3);

        // Show Books

        lib.showBooks();

        // Borrow Books

        System.out.println("\nBorrowing Books:");
        System.out.println(b1.borrowBook());
        System.out.println(b1.borrowBook()); // Try again

        // Return Book

        System.out.println("\nReturning Book:");
```

```
System.out.println(b1.returnBook());
```

```
// Show Updated Books
```

```
lib.showBooks();
```

```
// Create Users (Polymorphism)
```

```
User[] users = {
```

```
    new Student("Rahim"),
```

```
    new Teacher("Karim")
```

```
};
```

```
// Display Roles
```

```
Display.showRoles(users);
```

```
}
```

```
}
```